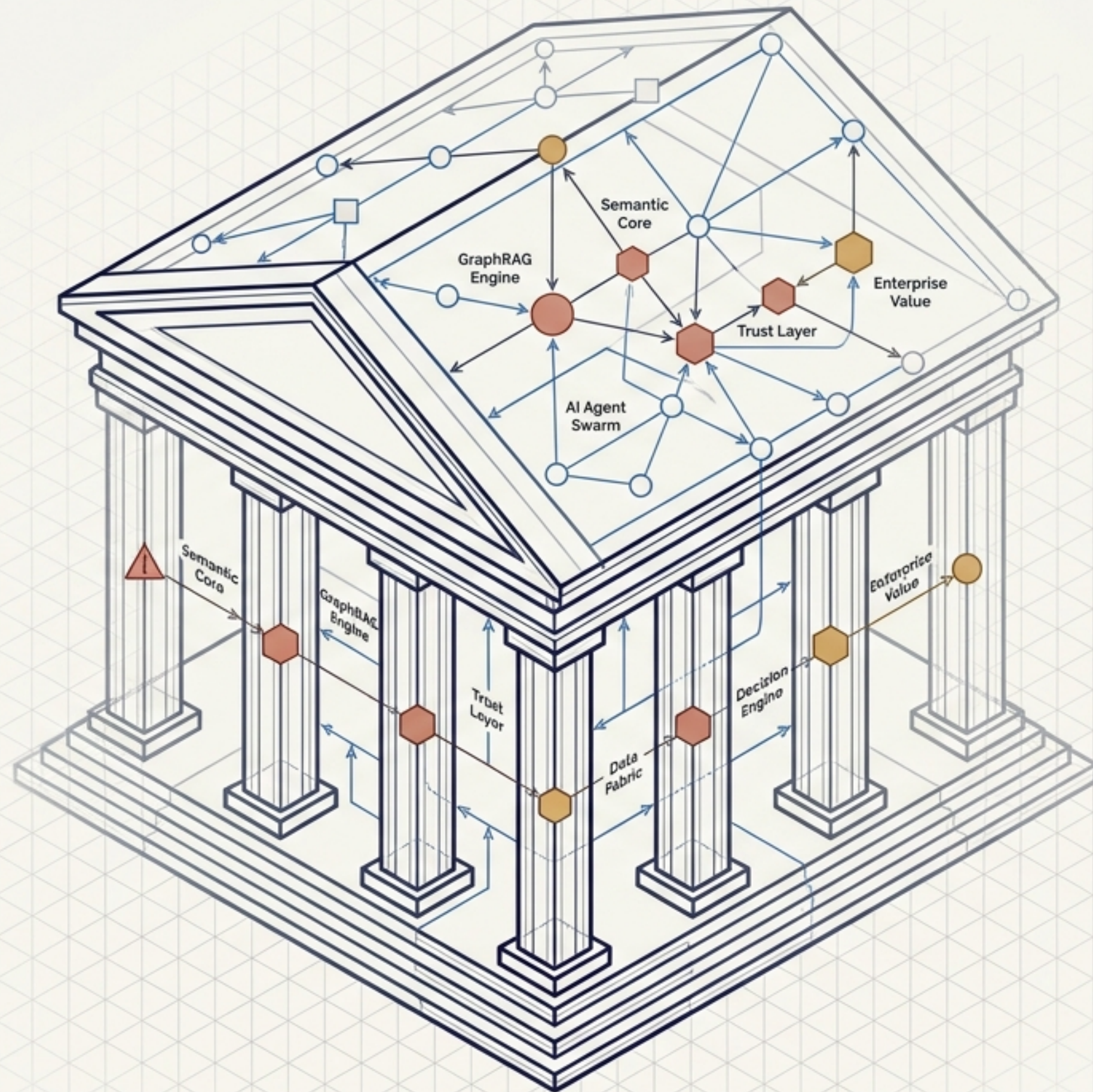


AGENTIC INTELLIGENCE AT SCALE

An Architectural Blueprint for GraphRAG,
Semantic Modeling, and Trustworthy AI.

Prepared for Enterprise Architecture & D&A Leadership



Aligning the Triad of Enterprise Intelligence

The transition from siloed, account-centric reporting to an automated, household-centric Agentic AI ecosystem requires a unified architectural mandate.

Data Strategy (The "How - Infrastructure")

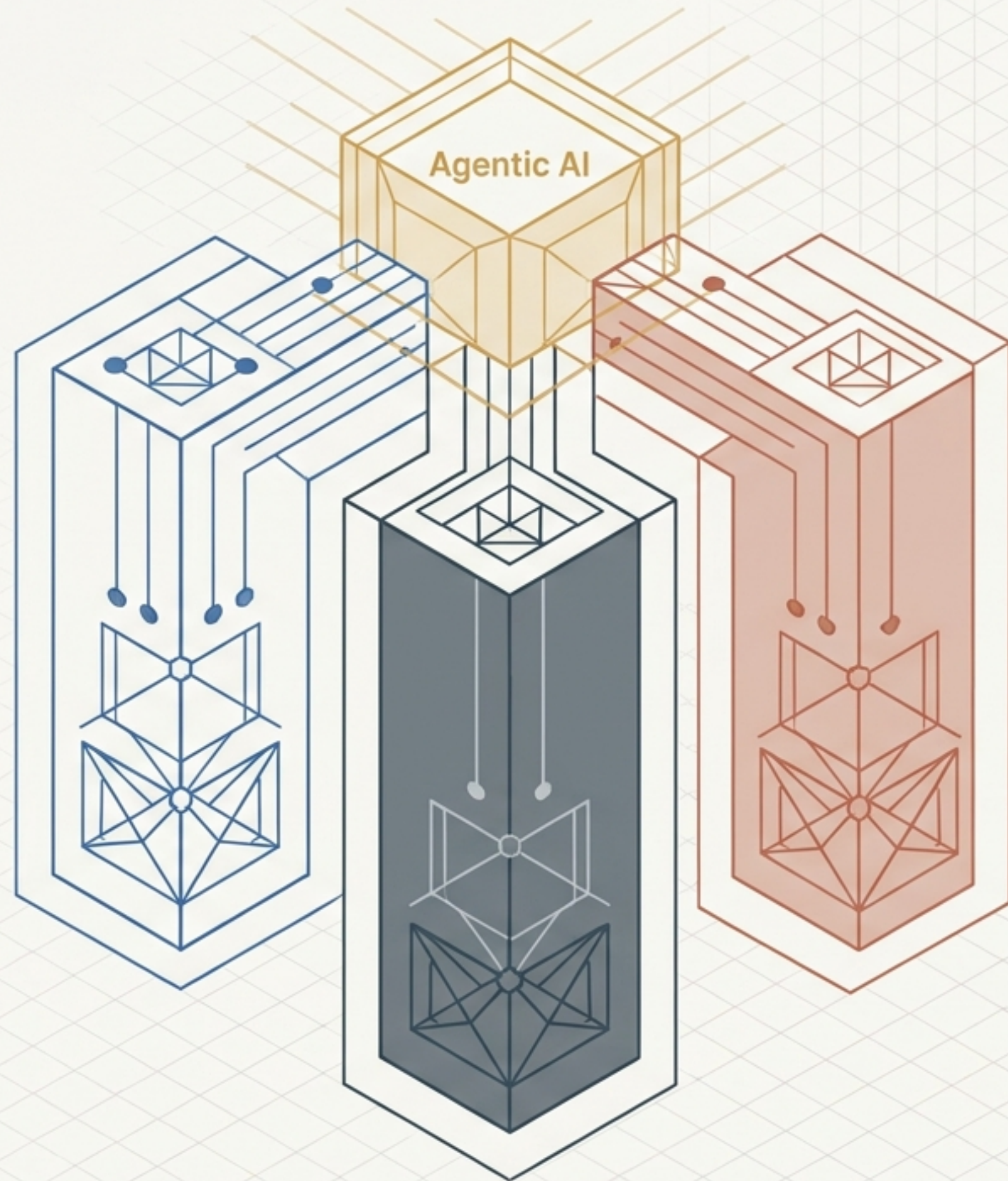
Decouple legacy silos into a governed cloud lakehouse using Data Contracts and Data Vault 2.0 principles.

Business Strategy (The "Why")

Automate complex financial decisioning (e.g., mortgage pre-approvals, fraud triage) by assessing multi-hop household ROI.

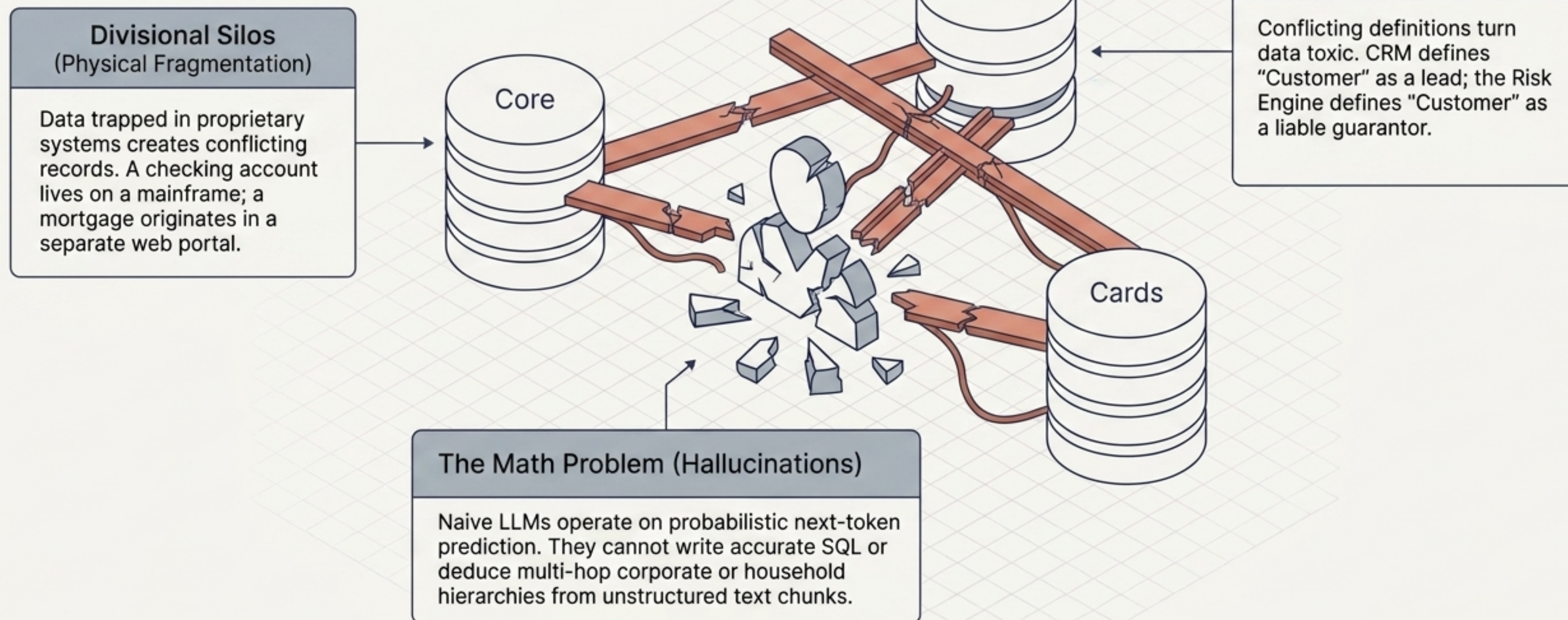
AI Strategy (The "How - Execution")

Replace probabilistic LLM guesses with deterministic, governed enterprise context via Graph-Semantic Grounding.

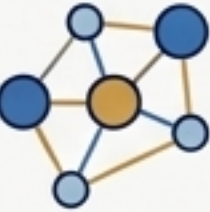


Why Naive AI Fails in the Enterprise

Centralizing data into a lakehouse without a semantic strategy simply creates a data swamp, leading directly to catastrophic LLM hallucinations.

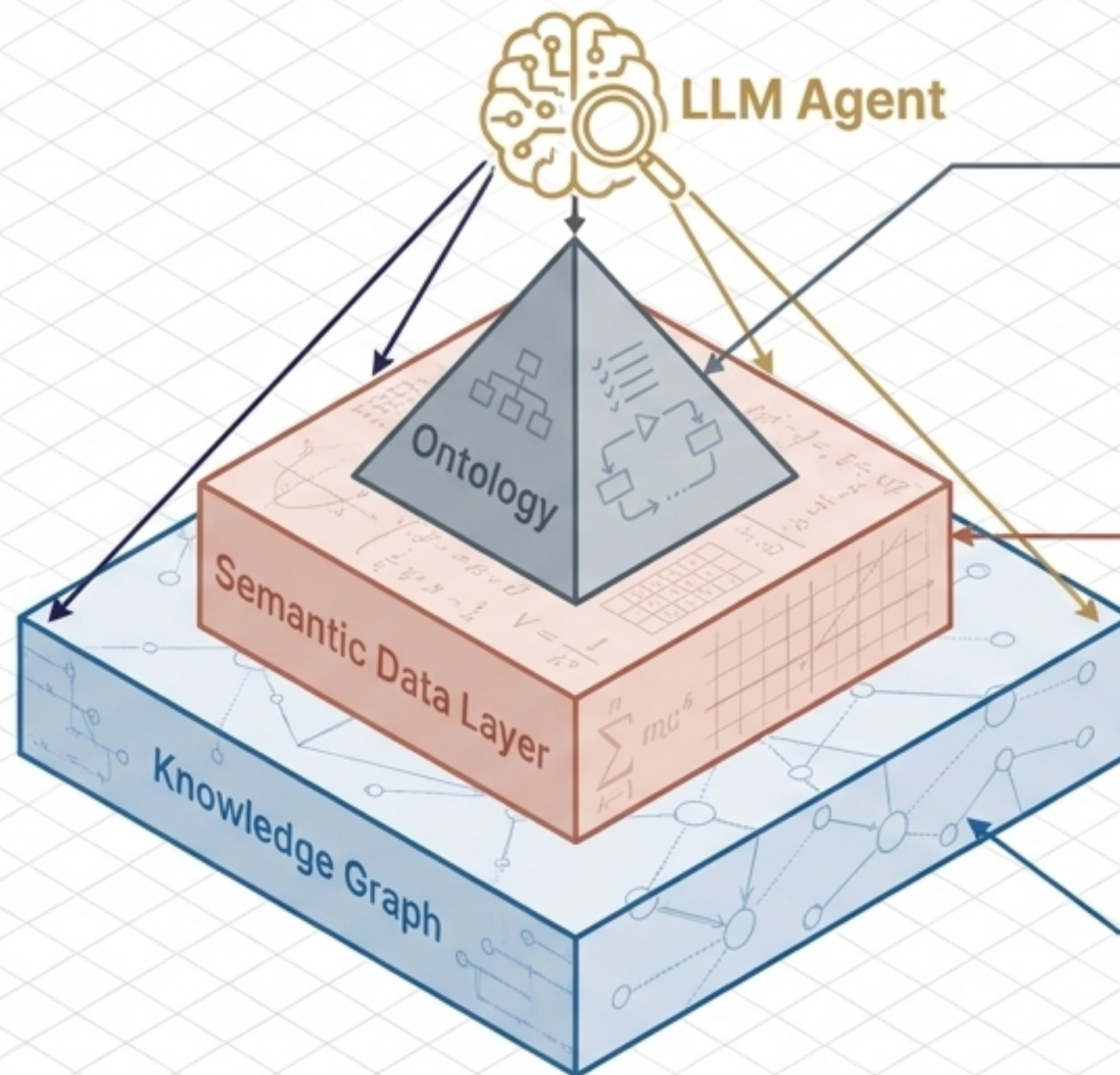


The Execution Shift: Vector RAG vs. Agentic GraphRAG

Capability Dimension	Naive Vector-Only RAG (Status Quo)	Agentic GraphRAG (Proposed)
Retrieval Mechanism	Top-K cosine similarity on unstructured text chunks. 	Sub-graph traversal over explicit, governed entities & relationships. 
Complex Math & Aggregations	LLM attempts to write raw SQL; frequently hallucinates logic. 	Executes pre-governed formulas via Model Context Protocol (MCP) APIs. 
Multi-Hop Reasoning	Fails entirely when facts span >2 disconnected documents. 	Instant 10+ hop traversal across connected Knowledge Graph nodes. 
Explainability & Lineage	Black-box vectors; no audit trail for regulatory compliance. 	Full lineage to specific graph edges and dbt MetricFlow YAML definitions. 

The Triad of Trust: Grounding the AI

To eliminate hallucinations, the LLM must act purely as a reasoning orchestrator, never as a calculator or database. It is constrained by three governed layers:



1. Ontology (The Rules)

Formal, machine-readable rules of the domain using W3C RDF/OWL. (e.g., A Guarantor is legally liable for a Credit Facility).

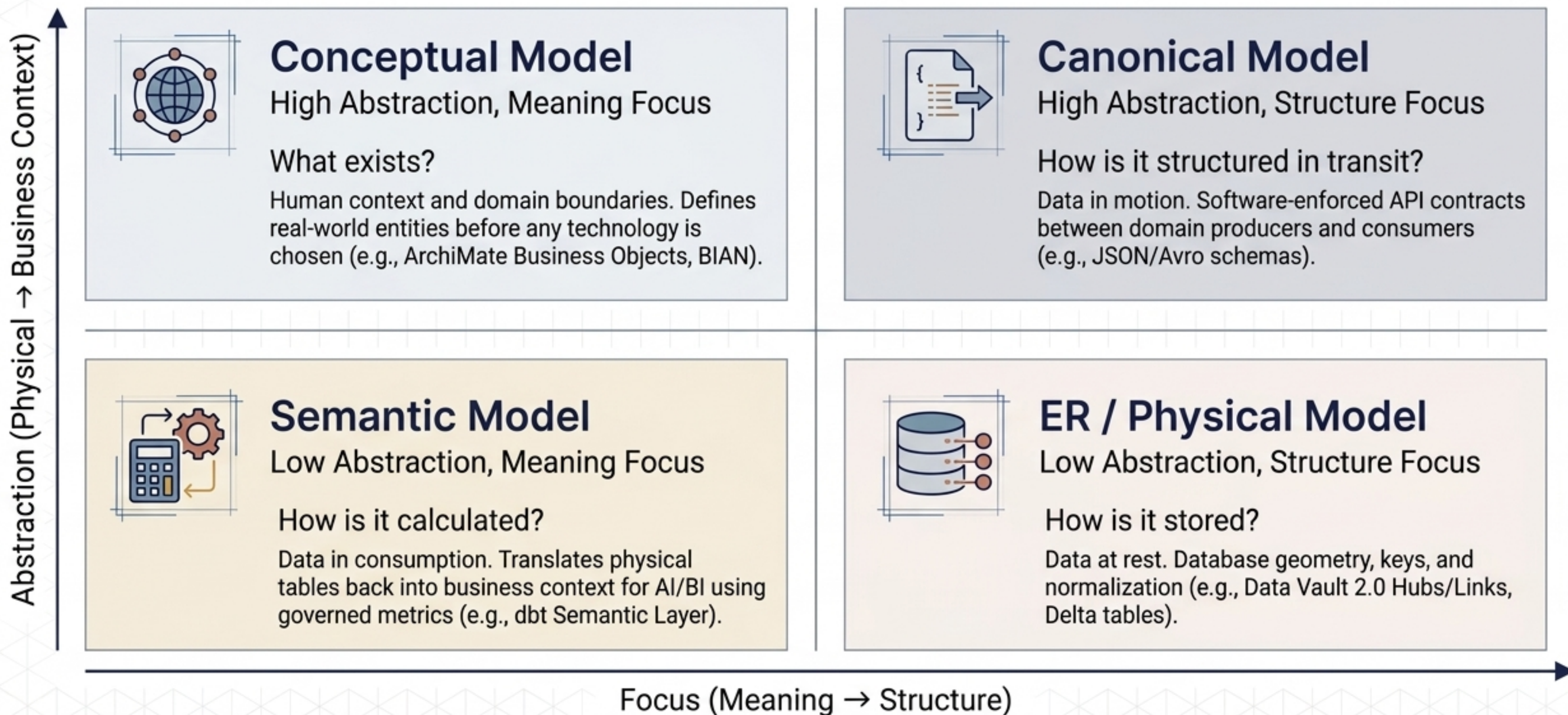
3. Semantic Data Layer (The Math)

Governed metrics and execution APIs built in **dbt MetricFlow**. Hardcodes complex math like Debt-to-Income (DTI) so the LLM never guesses the formula.

2. Knowledge Graph (The Facts)

Connected data nodes and edges providing explicit context (e.g., Jane Doe -> SHARES_ADDRESS_WITH -> John Doe).

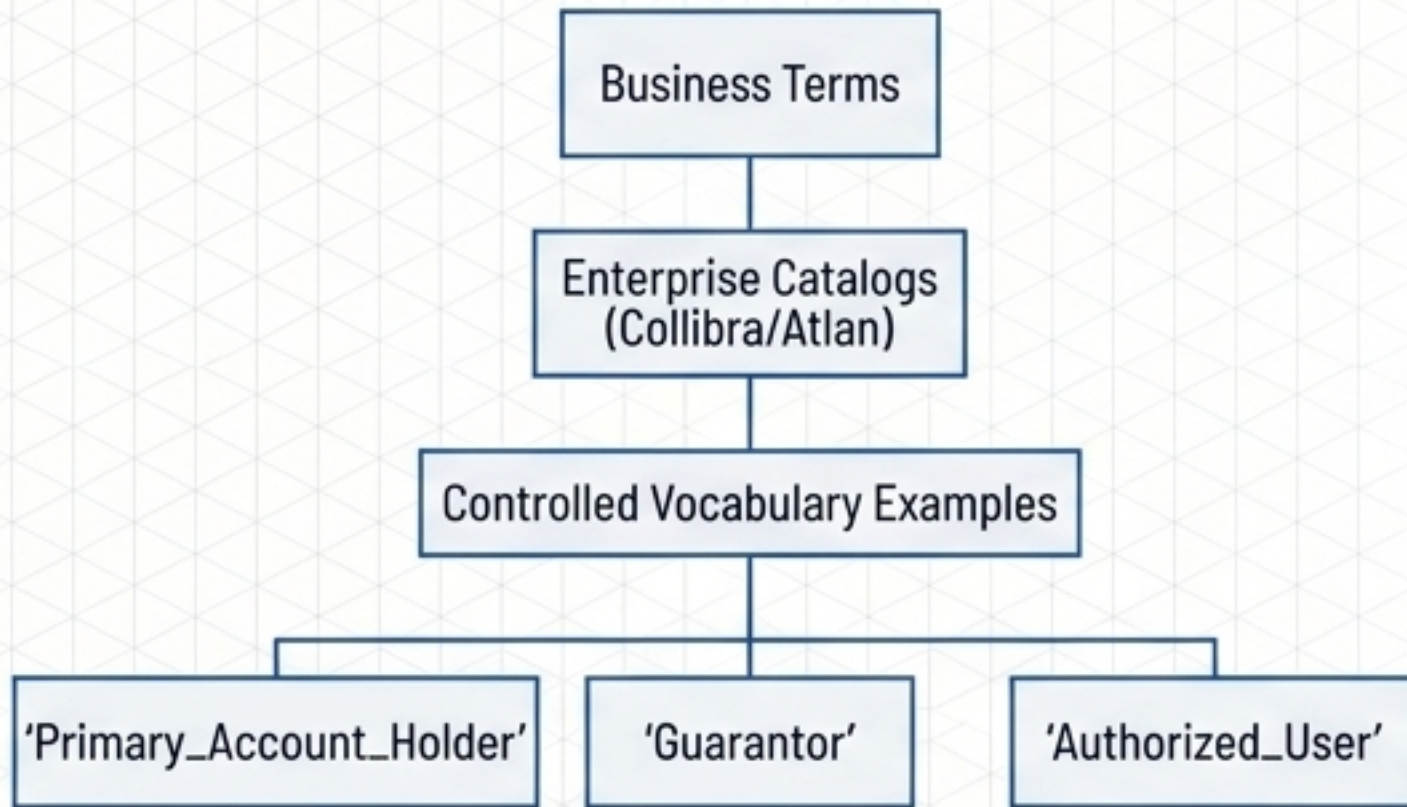
Decoding the Architectural Components



Information Architecture: Taxonomy & Ontology

Semantic ambiguity is resolved before a single line of SQL is written.

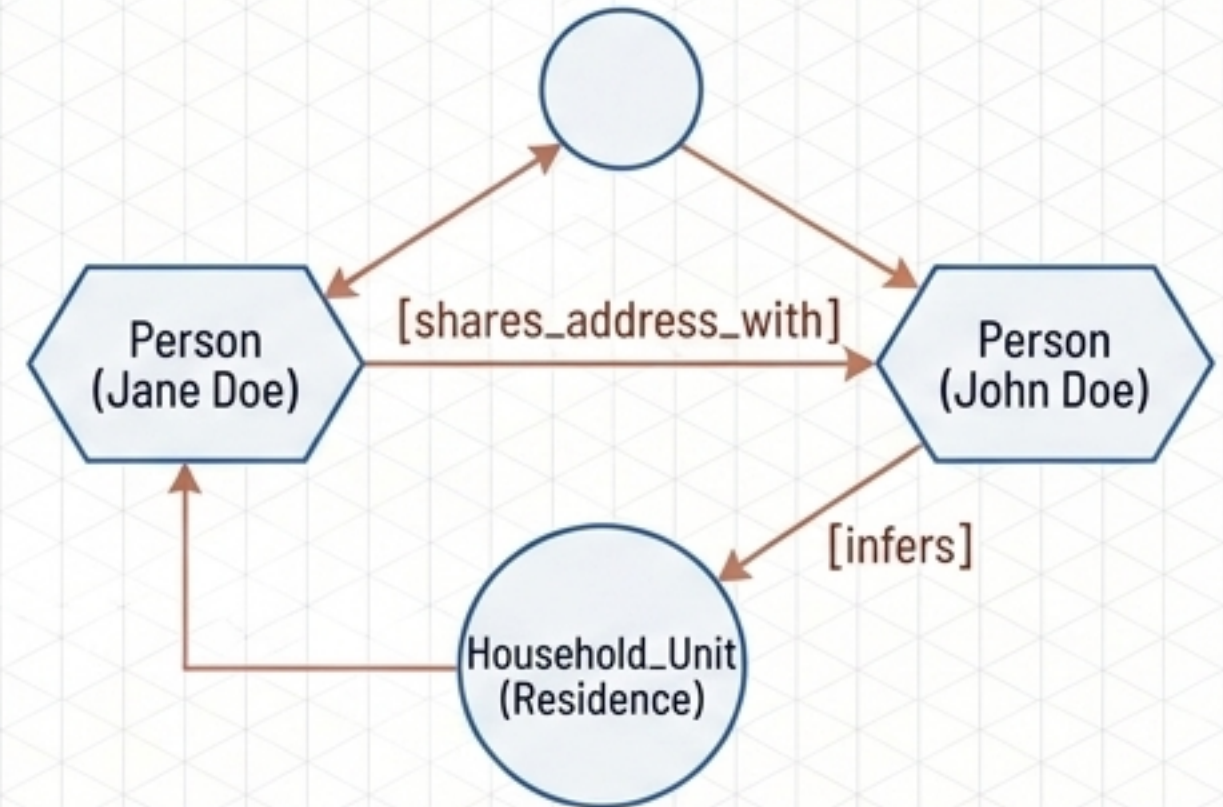
Taxonomy (Controlled Vocabulary)



Locking business terms in enterprise catalogs (Collibra/Atlan).

Example: Differentiating a 'Primary_Account_Holder' from a 'Guarantor' or an 'Authorized_User'.

Ontology & Conceptual Modeling



Defining formal axioms and real-world relationships.

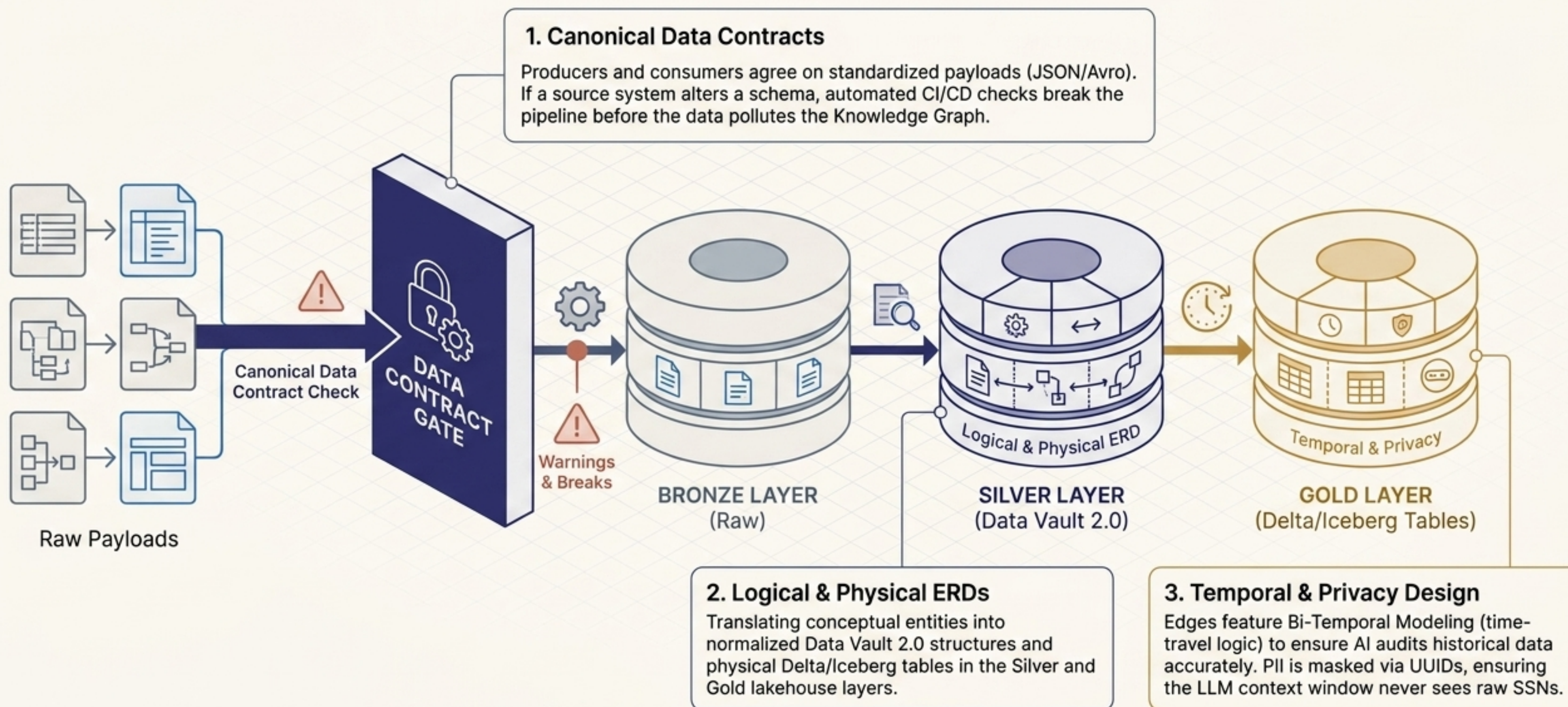
Axiom: Person [shares_address_with] Person [infers] Household_Unit.

Value:

Eliminates the 'Tower of Babel' problem. Ensures Risk, Marketing, and AI models all share a single, mathematically rigorous definition of reality.

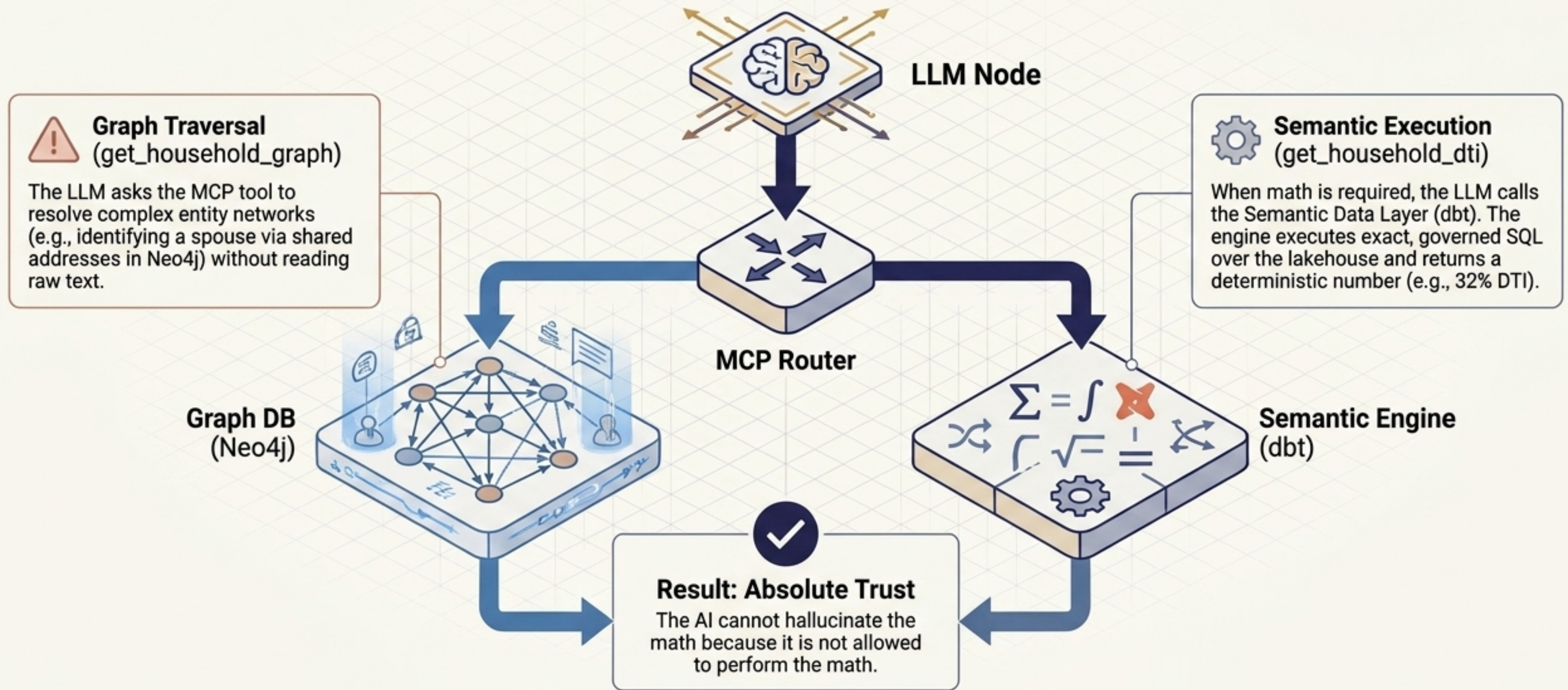
Data Architecture: Canonical Contracts & Lakehouse Storage

Eradicating $N \times M$ integration chaos through software-enforced boundaries.

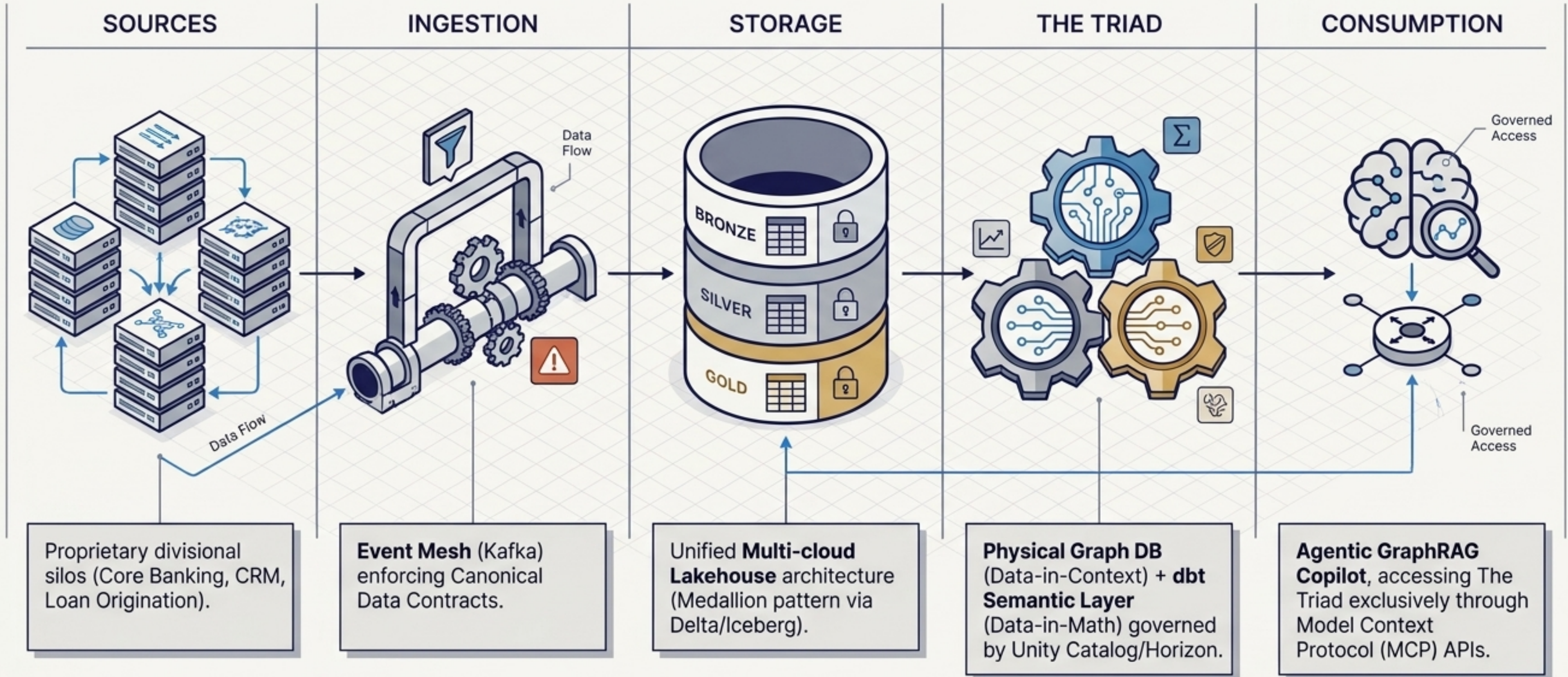


AI Execution: Model Context Protocol (MCP) & Tools

The LLM acts purely as a reasoning orchestrator. It accesses enterprise truth via strict, serverless APIs (MCP Tools).

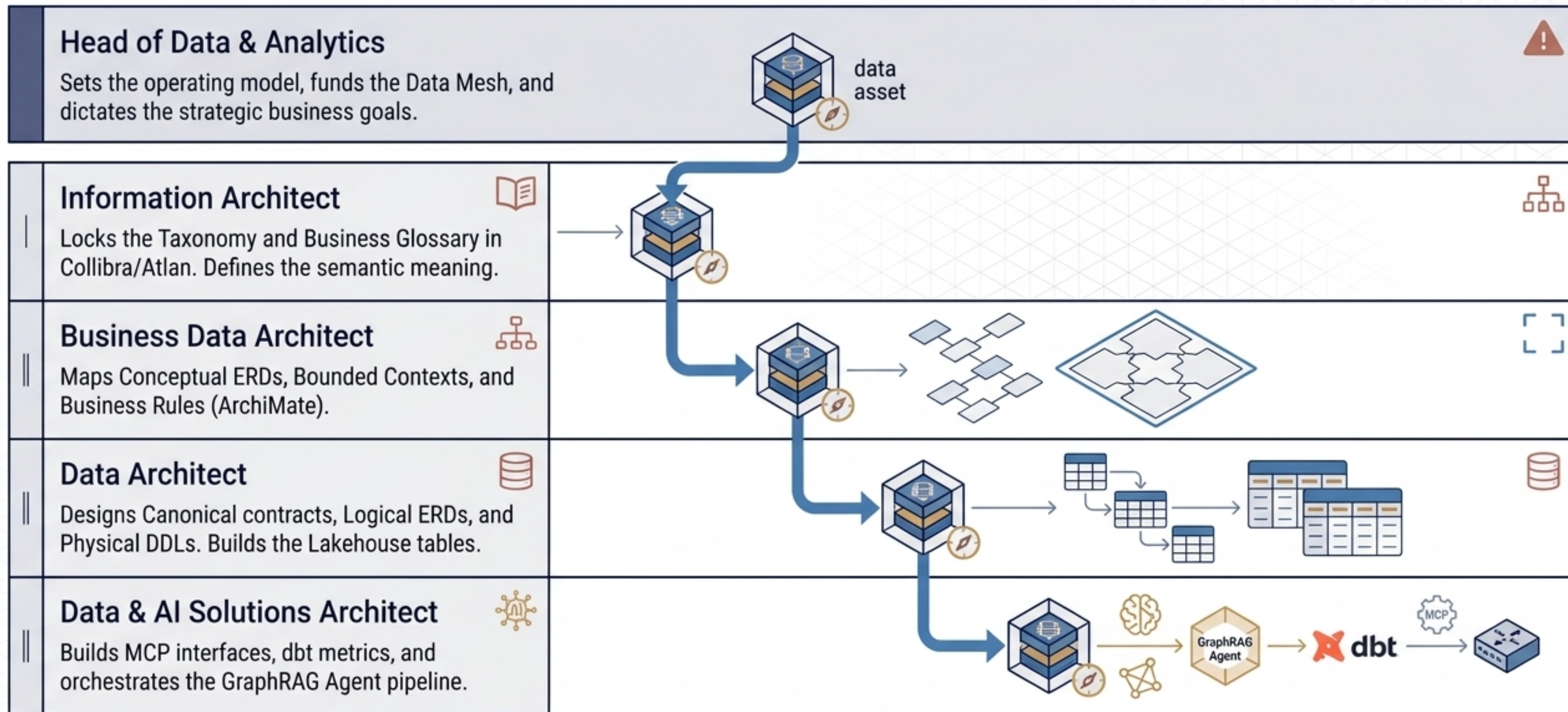


End-to-End Enterprise Architecture Flow



The Operating Model: Orchestrating the Bench

Transitioning strategy to execution requires a precise chain of handoffs.



Applied Use Case: Retail Mortgage Copilot

Traditional banking systems fail to assess the true financial health of modern, complex households.

The Scenario

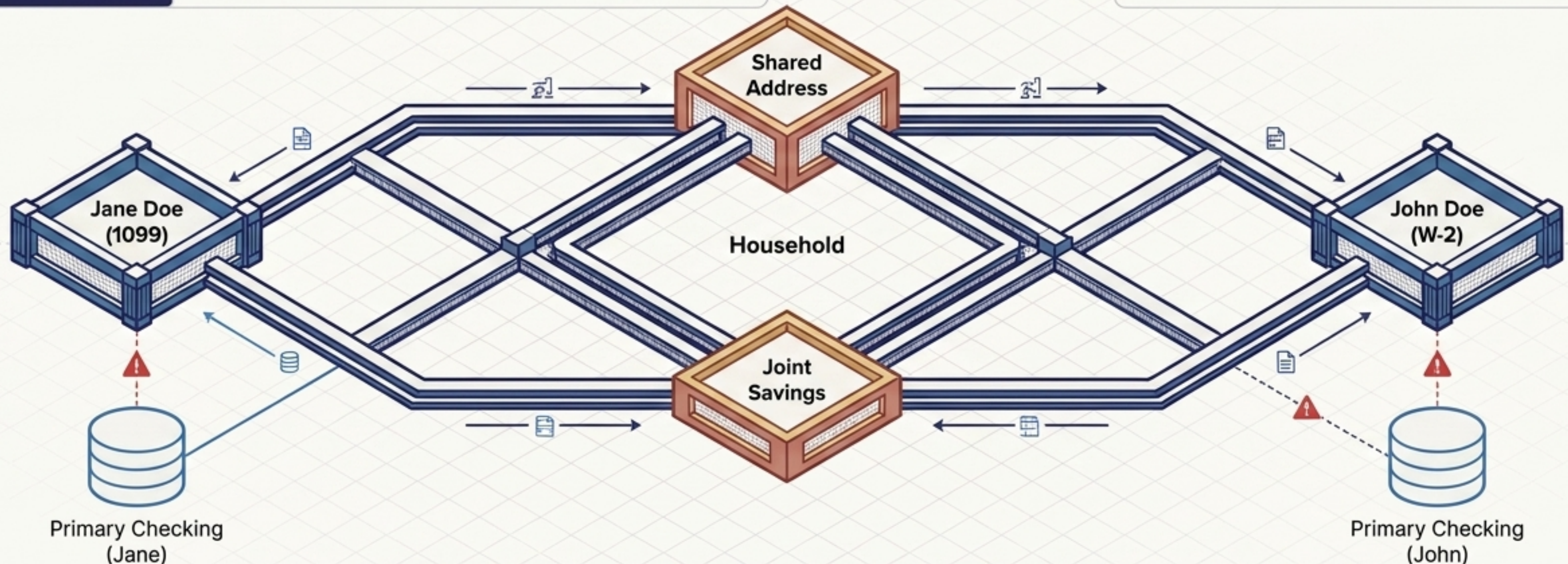
Assess complex mortgage affordability for a gig-economy worker by evaluating total household liquidity and Debt-to-Income (DTI).

The Challenge

Core banking systems see them as two isolated risk profiles; traditional underwriting rejects the application.

The Reality (Hidden Connections)

Jane Doe (1099 Freelance) and John Doe (W-2) maintain separate primary checking accounts, but are linked by a shared physical address and a joint savings account.



The Proof: Vector RAG vs. GraphRAG Execution

A stark split-screen diagnostic scenario walkthrough highlighting the impact of graph-based semantic reasoning.

Baseline (Naive Vector RAG) — FAILURE

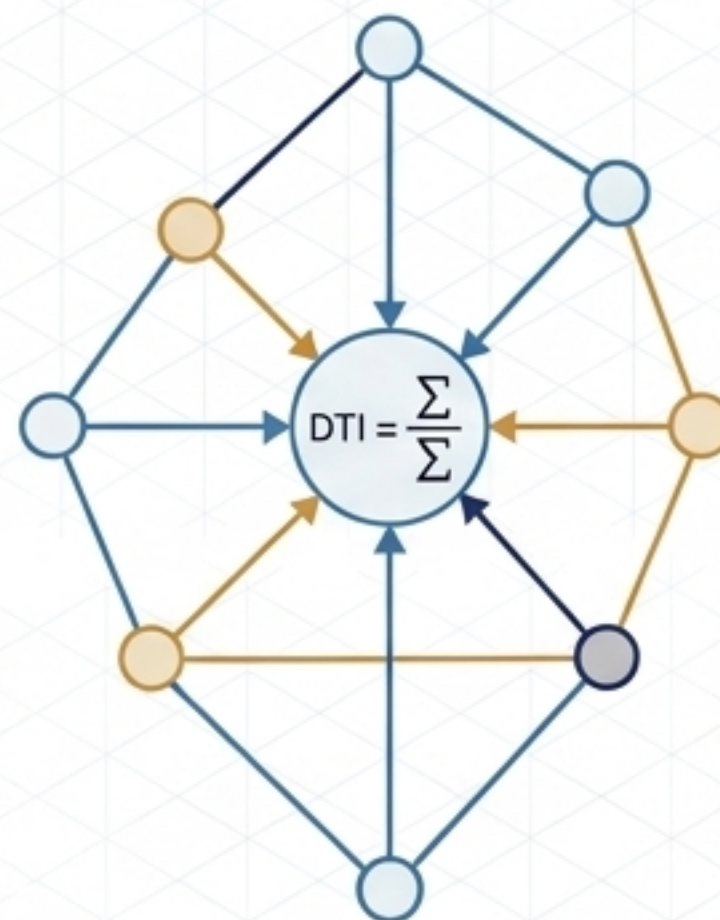
Action: LLM vector-searches checking statements.

Error: Misses the spouse entirely due to separate accounts. Hallucinates DTI math by dividing her joint car loan exclusively against her freelance income.

Result: Loan denied. High regulatory risk, lost revenue.



Proposed (Agentic GraphRAG) — SUCCESS



Action: MCP tool `get_household_graph` traverses Neo4j, linking Jane and John via shared address. MCP tool `get_household_dti` calls the dbt Semantic Layer for governed math.

Result: Combines John's W-2 income with Jane's 1099 income. Returns a deterministic 32% DTI. Agent safely approves a \$650k mortgage with zero hallucinations.

Value Creation & Enterprise ROI

Architectural rigor unlocks measurable, enterprise-wide business outcomes.

Revenue Growth



Unlock underserved segments (gig-workers, multi-generational households) by pricing risk on true multi-hop household liquidity.

Risk & Compliance



Eradicate LLM hallucinations. Full explainability via Semantic layer lineage. Flag synthetic first-party fraud instantly via network graph traversal.

Operational Efficiency



Reduce Level-1 alert triage and manual discovery from hours/days to minutes.

Strategic Roadmap & Next Steps

**Align the architecture.
Govern the semantics.
Unleash the AI.**

